DJANGO 6.1 • FIELD OPERATIONS RUNBOOK

Project Manager: Inventory Architecture & Walkthrough

Category Prefixes, Auto Part Numbering, Datasheets, Scans, Articles & Controller Matrix

HOST: **localhost:8000**
WORKSPACE: **Project_manager**
THEME: **Cadmium Green (#137547)**

1 COMPREHENSIVE CONTROLLER & ROUTE REGISTRY

30 ACTIVE ENDPOINTS

DOMAIN / MODEL	URL PATTERN	CLASS-BASED VIEW	TEMPLATE RESOURCE	ACTION
Dashboard	`/`	`views.home`	`templates/home.html`	KPI OVERVIEW
Part	`/parts/` | `/parts/add/`	`PartListView` | `PartCreateView`	`core/part_list.html` | `form.html`	SEARCH/CREATE
	`/parts/<pk>/`	`PartDetailView`	`core/part_detail.html`	STOCK SPECS
	`/parts/<pk>/edit/` | `/delete/`	`PartUpdateView` | `DeleteView`	`part_form.html` | `confirm_delete.html`	EDIT / PURGE
Documents & Scans	`/parts/<part_pk>/documents/add/`	`PartDocumentCreateView`	`core/part_document_form.html`	ATTACH FILE
	`/documents/<pk>/` | `/edit/` | `/delete/`	`DetailView` | `UpdateView` | `DeleteView`	`part_document_detail.html`	HIGH-RES VIEW
Category	`/categories/` | `/add/` | `/<pk>/`	`CategoryListView` | `CreateView` | `DetailView`	`category_list.html` | `detail.html`	TAXONOMY
Company	`/companies/` | `/add/` | `/<pk>/`	`CompanyListView` | `CreateView` | `DetailView`	`company_list.html` | `detail.html`	SUPPLIERS
Manufacturer	`/manufacturers/` | `/add/` | `/<pk>/`	`ManufacturerListView` | `CreateView`	`manufacturer_list.html`	OEM INDEX
Location	`/locations/` | `/add/` | `/<pk>/`	`LocationListView` | `CreateView`	`location_list.html`	STORAGE BINS

2 CATEGORY PREFIX & TAXONOMY

CORE/MODELS.PY

- **Prefix Field**: `Category.prefix` (e.g. `100` for Passives, `200` for Discrete Semiconductors, `300` for Connectors).
- **Recursive Inheritance**: `get_prefix()` checks current category; if empty, ascends parent tree (`parent.get_prefix()`).
- **Dropdown Display**: `Category.__str__()` formats hierarchies: `[100] Passives > Resistors > SMD 0805`.
- **Taxonomy Nesting**: Self-referential `parent = ForeignKey('self')` allows unlimited subcategory depth.

Subcategory Inheritance: Defining `100` on *Passives* allows *Resistors* and *Capacitors* to inherit prefix `100` without re-typing.

3 AUTOMATED PART NUMBERING

PREFIX ENGINE

- **Starting Sequence**: Sequential numbers start at `1001` (e.g. `100-1001`, `100-1002`, ..., `100-1234`).
- **Dynamic Increment**: `generate_next_part_number(category)` queries existing parts with `prefix-*`, parses numeric suffixes, and takes `max_num + 1`.
- **Intelligent Formatting in `save()`**:
  - Blank → Auto-generates next available sequence (`100-1001`).
  - Digits only (e.g. `1234`) → Prepends prefix: `100-1234`.
  - Custom string → Preserved untouched.
- **Collision Avoidance**: Loops candidate numbers until verified unused.

4 PART DOCUMENTS, SCANS, PINOUTS & TECHNICAL ARTICLES

PARTDOCUMENT MODEL

- **Relationship**: One-to-Many with `Part` (`part.documents.all()`). Multiple attachments per part number.
- **Document Types**: `datasheet` (PDF) • `pinout` (Diagram/Schematic) • `scan` (Document Scan) • `instructions` (Assembly Guide) • `article` (Technical Notes)
- **File Storage**: Uploads to `media/parts/documents/%Y/%m/`. Auto-detects `is_image`, `is_pdf`, and formats file size.
- **External Links**: `external_url` field for manufacturer-hosted online datasheets and web references.
- **WYSIWYG Articles**: `content = CKEditor5Field()` for rich pinout tables, troubleshooting procedures, and markdown notes.
- **Dedicated Viewers**: Embedded in-browser PDF reader, high-resolution scan zoom canvas, and print-ready technical article viewer.

RESOURCE KIND	TYPICAL FILES / CONTENT	FRONTEND BEHAVIOR ON PART DETAIL
Datasheet	Official PDF datasheet or direct web link	Red PDF badge, file size indicator, embedded viewer, download link
Pinout Diagram	PNG / JPG schematic scan + pin table	Visual thumbnail preview, image zoom modal, pinout table
Lab Scan	Handwritten notebook scan or CAD drawing	High-res image canvas with full resolution popout
Instructions	Rich WYSIWYG text with formatting & steps	Article reader view with formatted tables, bold text, code blocks

5 SOURCING & LOCATION TOPOLOGY

ECOSYSTEM

- **Company (Supplier)**: Vendor supplying components (DigiKey, Mouser). Stores account numbers, sales reps, and phone contacts.
- **Manufacturer (OEM)**: Original brand (Texas Instruments, STMicro).
- **Location**: Physical warehouse, rack, or bin code (`BIN-A12`).
- **Price Automation**: `price_per` × `quantity` → auto-computes `price` in model `save()`.
- **Superseding**: Self FK `superseded_by` links legacy parts to newer replacement inventory items.

6 THEME & UI ARCHITECTURE

CADMIUM GREEN

- **Base Palette**: Integrated `bootstrapcolors-custom.css`: Cadmium Green (`#137547`), Forest (`#054a29`), Mint accents.
- **Responsive Sidebar**: Permanent desktop drawer; collapses into touch-friendly offcanvas on mobile devices.
- **Dual Dark/Light Theme**: Bootstrap 5.3 `data-bs-theme` with persistent client-side `localStorage` sync.
- **Interactive Forms**: Real-time JS total cost calculator & dynamic category prefix placeholders.

7 OPERATOR WALKTHROUGH & TERMINAL RUNBOOK

STANDARD PROCEDURE

```
# 1. Boot local development server source .venv/bin/activate python manage.py runserver 127.0.0.1:8000 # 2. Create Category with Prefix (e.g. 100 for Passives) curl -X POST http://127.0.0.1:8000/categories/add/ -d "name=Passives&prefix=100&description=Passive Components" # 3. Create Part leaving part_number blank -> auto-generates 100-1001 curl -X POST http://127.0.0.1:8000/parts/add/ -d "description=10uF Capacitor&category=1&quantity=100&price_per=0.15" # -> Generated: part_number = '100-1001' | price = 15.00 # 4. Create Part entering specific sequence '1234' -> auto-formats to 100-1234 curl -X POST http://127.0.0.1:8000/parts/add/ -d "part_number=1234&category=1&description=Precision Resistor" # -> Formatted: part_number = '100-1234' # 5. Attach a Datasheet / Scan to the component curl -X POST http://127.0.0.1:8000/parts/1/documents/add/ \ -F "title=TI LM317 Datasheet" \ -F "document_type=datasheet" \ -F "file=@datasheet.pdf"
```

OPERATIONAL PHASE	ACTION PERFORMED IN INTERFACE	SYSTEM REACTION & STATE CHANGE
1. Category Setup	Go to `/categories/add/`. Enter Name & Prefix (e.g. `100`).	Prefix stored; inherited by any child subcategories.
2. Part Registration	Select category on `/parts/add/`. Leave part number blank.	Auto-generates `100-1001`. Computes `price = price_per * qty`.
3. Suffix Override	Enter `1234` in Part Number input.	Auto-prepends category prefix to produce `100-1234`.
4. Attach Resources	On Part detail page, click *Add Document / Scan*.	Uploads PDF/scan, renders thumbnail, and attaches to part record.
5. Document Viewing	Click *View* on any document card.	Opens dedicated high-res viewer, embedded PDF, or article body.